GATE 2027

General Aptitude

Topic-wise Question Bank

Common to the CSE and DA papers

222 questions • **16** topic sets • **4** official syllabus sections

181 practice questions + **41** real GATE previous-year questions

MCQ • MSQ • NAT — the three GATE question types

One combined answer key at the very end

Where every question comes from — read this first. This book has two parts, and they are labelled differently on purpose.

Part A (Q.1–Q.181) — practice questions written for this book. Each carries a *Type / syllabus* line naming the official syllabus topic it tests and the recurring GA form it follows. These are **not** past-paper questions and are never labelled as such.

Part B (Q.182–Q.222) — 41 genuine previous-year questions, marked **PYQ** and grouped under the same topic sets. Every one is reproduced from the **official question paper** published by the GATE organising institute, and its answer is the **official answer key** for that paper. The source line under each names the exact paper and question number, e.g. *GATE CS 2024 (Set 1), Q.4*. Papers used: GATE DA 2025 and 2024; GATE CS 2025 (Set 1), 2024 (Set 1), 2023 and 2022.

Every numerical answer in Part A was recomputed independently before the key was fixed, and every Part B answer that can be derived was re-derived and checked against the official key — all agreed.

Prepared for Ashutosh • August 2026

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Answer Key **56**

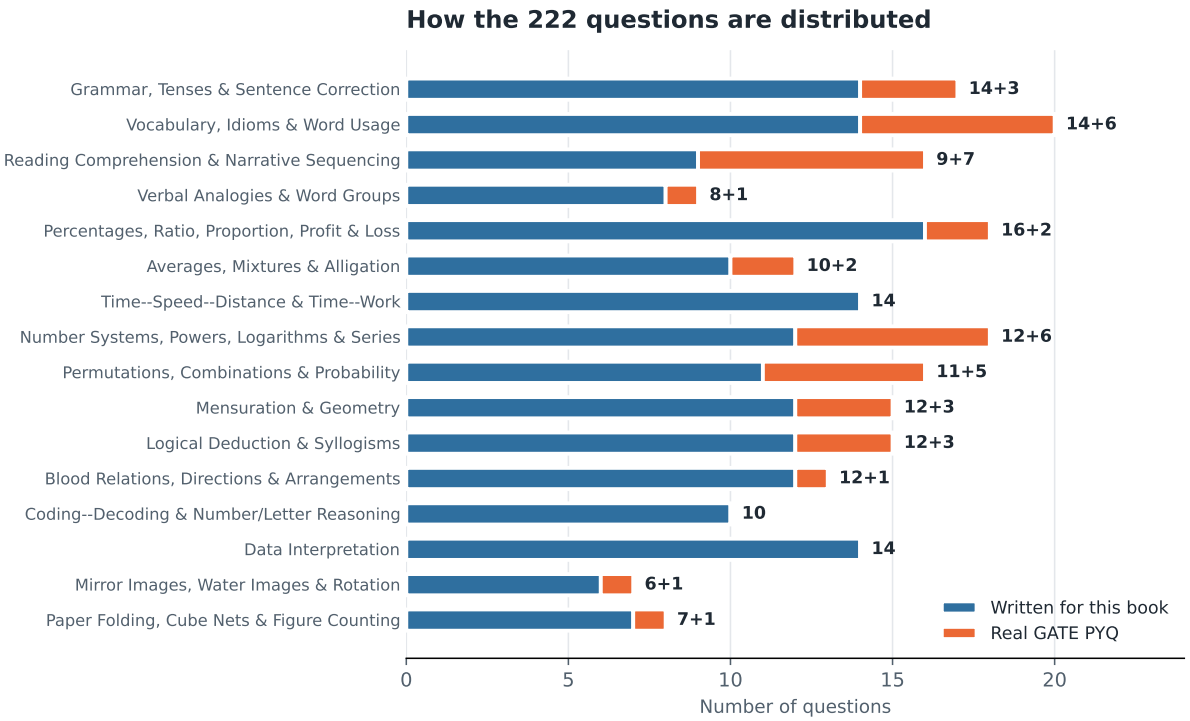
How this book is built

General Aptitude is the same 15 marks in every GATE paper. In GATE CSE and GATE DA alike it is **Q.1–Q.10**: five one-mark questions and five two-mark questions, 15 of the 100 marks. It has been exactly this in every recent paper, including all three GATE DA papers (2024, 2025, 2026). Unlike the technical subjects — where the marks swing by eight or more between years — **GA does not move**. That makes it the most predictable block in the paper and the cheapest marks available to you.

The official GATE 2027 GA syllabus has four sections: Verbal Aptitude (grammar — tenses, articles, prepositions, conjunctions, verb–noun agreement; vocabulary — words, idioms and phrases in context; reading and comprehension; narrative sequencing); **Quantitative Aptitude** (data interpretation; numerical computation and estimation — ratios, percentages, powers, exponents and logarithms, permutations and combinations, series; mensuration and geometry; elementary statistics and probability); **Analytical Aptitude** (logic — deduction and induction; analogy; numerical relations and reasoning); and **Spatial Aptitude** (transformation of shapes — translation, rotation, scaling, mirroring, assembling and grouping; paper folding, cutting and patterns in two and three dimensions).

Why the topic sets are the sizes they are

The four official sections are not equally represented in the actual papers. Across GATE CSE and DA, the GA questions that appear are overwhelmingly **quantitative and verbal**, with analytical reasoning close behind and **spatial aptitude appearing only occasionally** — it is fully examinable but historically rare in these two papers. The question counts follow that observed reality rather than giving the four sections equal space. The orange part of each bar is the real previous-year questions in that topic set.



Official section	Questions	Share	Why this size
Quantitative Aptitude	93	42%	the largest and most reliable source of GA questions
Analytical Aptitude	52	23%	logic and DI appear in almost every paper
Verbal Aptitude	62	28%	3–4 of the 10 GA questions are typically verbal
Spatial Aptitude	15	7%	examinable but rare in CSE/DA — covered, not over-weighted
Total	**222**	**100%**	

A note on the previous-year questions

The 41 PYQs in Part B are every General Aptitude question from six recent official papers that can stand on its own in print. **Questions that depend on a figure, map, pie chart or dice net were left out** — reproducing those diagrams faithfully was not possible here, and a mis-drawn figure is worse than no question. That is why three topic sets (Time–Speed–Distance, Coding–Decoding, and Data Interpretation) carry no PYQ: in these six papers, the questions of those types were all figure-based. Practise those three sets from Part A and from the official papers directly.

Question types and marking

Type	Description	Negative marking
MCQ	four choices, exactly one correct	$-1/3$ (1-mark), $-2/3$ (2-mark)
MSQ	four choices, one or more correct; no partial credit	none
NAT	a numerical value entered directly	none

MSQs have grown in the DA paper from 10 to 12 to 14 questions across 2024–2026, so a proportionate share of Part A is in MSQ form. Treat every MSQ option as an independent true/false judgement — that is the skill it tests, and it is different from eliminating three wrong choices.

How to use it

1. Work one topic set of Part A at a time, timed at **75 seconds** per question. GA questions you cannot start within 20 seconds are questions to skip in the real exam.
2. Then do the matching Part B set for that same topic. These are the real thing at the real difficulty — treat your score there as the honest signal.
3. Attempt every NAT and every MSQ — neither carries a penalty. On MCQs, guess only after eliminating at least two options.
4. Aim to finish all ten GA questions in the real paper in **under 18 minutes**. Doing GA first protects it from being squeezed at the end.

PART A — Practice questions, topic-wise

Written for this book. Each question carries a *Type / syllabus* line naming the official syllabus topic and the recurring GA form it follows. These are not past-paper questions.

Section: Verbal Aptitude

V1. Grammar, Tenses & Sentence Correction

14 questions • **+3 PYQ in Part B** • Official syllabus: Verbal Aptitude → tenses, articles, prepositions, conjunctions, verb-noun agreement, parts of speech

Q.1 [MCQ • Easy]

Select the most appropriate option.

Neither of the two candidates _____ eligible for the post.

(A) is (B) have been (C) were (D) are

Type / syllabus: Verb–noun agreement (*neither of* takes a singular verb) — recurring GA Q.1–Q.5 type

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Q.2 [MCQ • Easy]

Select the most appropriate option.

The number of applicants _____ risen sharply this year.

(A) have (B) has (C) are (D) were

Type / syllabus: Verb–noun agreement (*the number of* is singular) — classic GA distractor pair

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Q.3 [MCQ • Easy]

Select the most appropriate article to complete the sentence.

He is _____ honest officer who has never accepted a favour.

(A) no article is needed (B) a (C) the (D) an

Type / syllabus: Articles (vowel *sound*, not vowel letter) — GA 1-mark grammar slot

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Q.4 [MCQ • Easy]

Select the most appropriate preposition.

She has been working at this laboratory _____ 2019.

(A) from (B) during (C) since (D) for

Type / syllabus: Prepositions (*since* + point in time) — GA 1-mark grammar slot

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Q.5 [MCQ • Moderate]

Select the most appropriate option.

If I _____ in your position, I would accept the offer.

(A) am (B) had been (C) was (D) were

Type / syllabus: Tenses — subjunctive mood in a second conditional

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Q.6 [MCQ • Moderate]

Select the most appropriate option.

Hardly had the chairman finished speaking _____ the audience began to applaud.

(A) then (B) when (C) that (D) than

Type / syllabus: Conjunctions — the *hardly . . . when* pair; frequently paired with the *no sooner . . . than* trap

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Q.7 [MCQ • Moderate]

Select the most appropriate option.

No sooner did the bell ring _____ the students rushed out of the hall.

(A) than (B) then (C) as (D) when

Type / syllabus: Conjunctions — the *no sooner . . . than* pair

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Q.8 [MCQ • Moderate]

Select the most appropriate option.

By the time you reach the office, we _____ the report.

(A) finished (B) will finish (C) have finished (D) will have finished

Type / syllabus: Tenses — future perfect for an action completed before a future reference point

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Q.9 [MCQ • Moderate]

Which one of the following sentences is grammatically **correct**?

(A) Each of the students are submitting an assignment.

(B) Each of the students have submitted an assignment.

(C) Each of the students has submitted an assignment.

(D) Each of the student have submitted an assignment.

Type / syllabus: Verb–noun agreement — *each of* + plural noun + singular verb

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Q.10 [MCQ • Moderate]

Select the option that completes the sentence without a dangling modifier.

Walking along the river bank, _____

(A) a heron was seen taking flight.

(B) we saw a heron take flight.

(C) the heron's flight was seen by us.

(D) there was a heron seen taking flight.

Type / syllabus: Parts of speech / modifier placement — the modifier must attach to the subject that performs the action

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Q.11 [MCQ • Moderate]

Select the most appropriate option.

The committee _____ divided in its opinion on the proposal.

(A) are being (B) were (C) was (D) have

Type / syllabus: Collective nouns with a singular possessive (*its*) take a singular verb

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Q.12 [MSQ • Moderate]

Which of the following sentence(s) is/are grammatically **correct**?

(A) The data support the conclusion drawn by the authors.

(B) Neither the manager nor the clerks were informed.

(C) One of my friends is joining us tomorrow.

(D) He is one of those students who is always punctual.

Type / syllabus: Verb–noun agreement across several structures — MSQ form, now the fastest-growing GA question type

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Q.13 [MCQ • Hard]

Select the option that maintains parallel structure.

She enjoys reading, writing and _____.

(A) painting (B) to paint (C) the painting (D) she paints

Type / syllabus: Parts of speech — parallelism in a list of gerunds

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Q.14 [MCQ • Hard]

Convert to reported speech: *He said, "I went to Delhi yesterday."*

(A) He said that he goes to Delhi the day before.

(B) He said that he went to Delhi yesterday.

(C) He said that he had gone to Delhi the previous day.

(D) He said that he has gone to Delhi the previous day.

Type / syllabus: Tenses — backshift and time-reference change in reported speech

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V2. Vocabulary, Idioms & Word Usage

14 questions • **+6 PYQ in Part B** • Official syllabus: Verbal Aptitude → basic vocabulary: words, idioms and phrases in context

Q.15 [MCQ • Easy]

Select the word that is closest in meaning to **UBIQUITOUS**.

(A) Ambiguous (B) Unique (C) Omnipresent (D) Rare

Type / syllabus: Basic vocabulary — synonym; GA Q.1–Q.5 vocabulary slot

Q.16 [MCQ • Easy]

Select the word that is most nearly **opposite** in meaning to **FRUGAL**.

(A) Thrifty (B) Extravagant (C) Fertile (D) Modest

Type / syllabus: Basic vocabulary — antonym; note the *frugal/fertile* sound-alike distractor

Q.17 [MCQ • Easy]

The idiom *"to beat around the bush"* means

(A) to avoid coming to the main point (B) to work in the open air

(C) to search very thoroughly (D) to defeat someone easily

Type / syllabus: Idioms and phrases in context — GA 1-mark slot

Q.18 [MCQ • Easy]

Select the most appropriate word.

The evidence presented was _____, so the committee reached a decision quickly.

(A) conclusive (B) conclusively (C) conclusion (D) concluding

Type / syllabus: Words and phrases — correct part of speech in context

Q.19 [MCQ • Moderate]

Select the word closest in meaning to **METICULOUS**.

(A) Scrupulous (B) Sceptical (C) Hasty (D) Ambitious

Type / syllabus: Basic vocabulary — synonym at GATE's usual difficulty band

Q.20 [MCQ • Moderate]

The idiom *"a blessing in disguise"* refers to

(A) a hidden threat (B) a gift given anonymously

(C) an apparent misfortune that later proves beneficial (D) a reward that is never delivered

Type / syllabus: Idioms and phrases in context

Q.21 [MCQ • Moderate]

Select the correct word.

The new tariff policy will _____ the cost of imported components.

(A) efficient (B) affect (C) affective (D) effect

Type / syllabus: Words and phrases — the *affect/effect* confusable pair, a standing GA favourite

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Q.22 [MCQ • Moderate]

Select the correct word.

She refused the bribe; for her it was a matter of _____.

(A) principle (B) principality (C) principal (D) principally

Type / syllabus: Words and phrases — the *principle/principal* confusable pair

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Q.23 [MCQ • Moderate]

Select the word most nearly **opposite** in meaning to **CANDID**.

(A) Evasive (B) Sincere (C) Frank (D) Cordial

Type / syllabus: Basic vocabulary — antonym

Q.24 [MSQ • Moderate]

Which of the following word(s) mean(s) *expressed in few words*?

(A) Succinct (B) Terse (C) Verbose (D) Concise

Type / syllabus: Word groups — MSQ form; tests a cluster of near-synonyms at once

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Q.25 [MCQ • Moderate]

EXACERBATE most nearly means

(A) to explain in detail (B) to soothe (C) to make worse (D) to accelerate

Type / syllabus: Basic vocabulary — synonym

Q.26 [MCQ • Hard]

To *"take something with a pinch of salt"* is to

(A) add detail to it (B) treat it as an insult

(C) accept it enthusiastically (D) regard it with scepticism

Type / syllabus: Idioms and phrases in context

Q.27 [MCQ • Hard]

LACONIC most nearly describes a person who

(A) is easily angered (B) is habitually late (C) speaks at great length (D) uses very few words

Type / syllabus: Basic vocabulary — less common word, upper end of the GA vocabulary band

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Q.28 [MCQ • Hard]

Select the word that does **not** belong with the others.

(A) Ecstatic (B) Jubilant (C) Elated (D) Morose

Type / syllabus: Word groups — odd-one-out on semantic field (three denote joy)

V3. Reading Comprehension & Narrative Sequencing

9 questions • **+7 PYQ in Part B** • Official syllabus: Verbal Aptitude → reading and comprehension, narrative sequencing

Q.29

[MCQ • Moderate]

Read the passage and answer the question that follows.

> *A dataset can be perfectly accurate and still be perfectly misleading. If a hospital records only the patients it admits, any model trained on those records learns about admission, not about illness. The measurement is correct; the population is wrong. This failure is quiet because it never shows up as an error in the data — every number in the table is exactly what it claims to be. It shows up only later, when the model is asked about people the hospital never saw.*

The central point of the passage is that

(A) hospitals routinely record their patient data inaccurately

(B) errors in a dataset are always visible in the table itself

(C) models should never be trained on hospital records

(D) a sample can be measured correctly and still not represent the population of interest

Type / syllabus: Reading and comprehension — main idea; GA 2-mark comprehension slot

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Q.30

[MCQ • Moderate]

With reference to Passage A above, the phrase *"the population is wrong"* implies that

(A) the study included people from too many different regions

(B) the number of people in the study is too small

(C) the group from which the data were drawn is not the group the conclusion is about

(D) the recorded values contain measurement errors

Type / syllabus: Reading and comprehension — vocabulary/phrase in context

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Q.31

[MCQ • Hard]

Which of the following can be **inferred** from Passage A?

(A) Larger datasets are always more reliable

(B) Checking a dataset for internal errors will not by itself reveal a sampling problem

(C) Illness and admission are unrelated

(D) Any model trained on hospital data will fail

Type / syllabus: Reading and comprehension — inference, not restatement

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Q.32

[MCQ • Moderate]

Read the passage and answer the question that follows.

> *For most of the twentieth century, the cost of storing a fact exceeded the cost of recomputing it, and software was written accordingly. That ratio has now inverted so completely that the opposite discipline is required: engineers must be talked out of keeping everything. The constraint is no longer the disk. It is that a stored fact must be kept true, and every copy of it is a promise that someone will maintain it.*

The author's main claim is that

(A) recomputing a fact is always cheaper than storing it

(B) storage has become so cheap that engineers should store everything

(C) the real cost of storing data is now the cost of keeping it correct

(D) twentieth-century software was badly written

Type / syllabus: Reading and comprehension — main idea

Q.33 [MCQ • Hard]

In Passage B, *"every copy of it is a promise"* is best understood as

(A) engineers should never duplicate data

(B) duplicated data is always more reliable

(C) copies of data are legally binding documents

(D) a duplicate creates an ongoing obligation to keep it consistent

Type / syllabus: Reading and comprehension — figurative phrase in context

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Q.34 [MCQ • Hard]

Which of the following would the author of Passage B most likely **agree** with?

(A) A design that stores less can be easier to keep correct

(B) Maintenance cost is independent of how much is stored

(C) Disk capacity is the main limit on modern systems

(D) Recomputation should always be preferred to storage

Type / syllabus: Reading and comprehension — author's viewpoint

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Q.35 [MCQ • Moderate]

The following sentences, when correctly ordered, form a coherent paragraph.

P. The result was a network that no single organisation owned.
Q. Early computer networks were built to link one laboratory to another.
R. Each new link was negotiated separately, machine by machine.
S. Only later did a common protocol make those negotiations unnecessary.

The correct order is

(A) Q R S P (B) Q S R P (C) P Q R S (D) R Q P S

Type / syllabus: Narrative sequencing — GA 2-mark slot; anchor on the sentence that needs no antecedent

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Q.36 [MCQ • Moderate]

Arrange the sentences into a coherent paragraph.

P. It therefore rewards the candidate who can decide quickly what *not* to attempt.
Q. The examination lasts three hours and contains sixty-five questions.
R. That leaves under three minutes for each question on average.
S. Some of those questions can take twice that long.

The correct order is

(A) Q R S P (B) Q S R P (C) R S Q P (D) Q R P S

Type / syllabus: Narrative sequencing — causal chain ending in the conclusion marker *therefore*

Q.37 [MCQ • Hard]

Arrange the sentences into a coherent paragraph.

P. Its accuracy on the training set was almost perfect.

Q. The team built a classifier to flag fraudulent transactions.

R. On live traffic, however, it flagged almost nothing.

S. The training data had contained one fraudulent case in every four.

The correct order is

(A) S Q P R (B) Q P R S (C) Q S P R (D) Q R P S

Type / syllabus: Narrative sequencing — the last sentence supplies the explanation for the contrast

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V4. Verbal Analogies & Word Groups

8 questions • **+1 PYQ in Part B** • Official syllabus: Verbal Aptitude → verbal analogies, word groups

Q.38 [MCQ • Easy]

Doctor : Hospital :: Teacher : _____

(A) Lesson (B) Student (C) School (D) Chalk

Type / syllabus: Verbal analogies — worker : workplace; GA Q.1–Q.2 analogy slot

Q.39 [MCQ • Easy]

Scarcity : Abundance :: Ascent : _____

(A) Climb (B) Summit (C) Slope (D) Descent

Type / syllabus: Verbal analogies — antonym relation

Q.40 [MCQ • Moderate]

Cobbler : Shoes :: Blacksmith : _____

(A) Horseshoes (B) Anvil (C) Coal (D) Forge

Type / syllabus: Verbal analogies — craftsman : product (note the tool distractors)

Q.41 [MCQ • Moderate]

Ornithology : Birds :: Entomology : _____

(A) Reptiles (B) Words (C) Insects (D) Fossils

Type / syllabus: Verbal analogies — discipline : object of study; *etymology* is the standard trap

Q.42 [MCQ • Moderate]

Kilogram : Mass :: Ampere : _____

(A) Voltage (B) Power (C) Electric current (D) Resistance

Type / syllabus: Verbal analogies — unit : quantity measured

Q.43 [MCQ • Moderate]

Sculptor : Statue :: Poet : _____

(A) Poem (B) Pen (C) Verse metre (D) Inspiration

Type / syllabus: Verbal analogies — creator : creation

Q.44 [MSQ • Hard]

In which of the following pair(s) does the second item stand to the first as a **tool stands to its user**?

(A) Carpenter : Chisel (B) Author : Novel (C) Surgeon : Scalpel (D) Farmer : Plough

Type / syllabus: Verbal analogies in MSQ form — separates *tool* from *product*

Q.45 [MCQ • Moderate]

Select the word that does **not** belong to the group.

(A) Triangle (B) Square (C) Circle (D) Pentagon

Type / syllabus: Word groups — odd-one-out; the other three are rectilinear polygons

Section: Quantitative Aptitude

Q1. Percentages, Ratio, Proportion, Profit & Loss

16 questions • **+2 PYQ in Part B** • Official syllabus: Quantitative Aptitude → numerical computation and estimation: ratios, percentages

Q.46 [MCQ • Easy]

25% of 480 is

(A) 120 (B) 140 (C) 115 (D) 96

Type / syllabus: Numerical computation & estimation → percentages, ratios — direct 1-mark computation

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Q.47 [NAT • Easy]

A number increased by 25% becomes 250. The original number is _____.

Type / syllabus: Numerical computation & estimation → percentages, ratios — reverse percentage; the common error is taking 75% of 250

Q.48 [MCQ • Easy]

Two quantities are in the ratio 3 : 5 and their sum is 64. The larger quantity is

(A) 24 (B) 45 (C) 40 (D) 32

Type / syllabus: Numerical computation & estimation → percentages, ratios — ratio partition

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Q.49 [NAT • Easy]

An article bought for Rs. 250 is sold for Rs. 300. The profit percentage is _____.

Type / syllabus: Numerical computation & estimation → percentages, ratios — profit and loss basics

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Q.50 [MCQ • Moderate]

The price of a commodity falls by 10% and then rises by 10%. The net change is

(A) a decrease of 2% (B) a decrease of 1% (C) no change (D) an increase of 1%

Type / syllabus: Numerical computation & estimation → percentages, ratios — successive percentage change; a perennial GA 2-mark trap

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Q.51 [NAT • Moderate]

A shopkeeper marks his goods 40% above cost price and then allows a discount of 25%. His profit percentage is _____.

Type / syllabus: Numerical computation & estimation → percentages, ratios — markup with discount

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Q.52 [MCQ • Moderate]

If $A : B = 2 : 3$ and $B : C = 4 : 5$, then $A : C$ equals

(A) 2 : 5 (B) 6 : 20 (C) 8 : 15 (D) 3 : 4

Type / syllabus: Numerical computation & estimation → percentages, ratios — chained ratios

Q.53 [NAT • Moderate]

In an examination 60% of the candidates passed and 120 candidates failed. The total number of candidates is _____.

Type / syllabus: Numerical computation & estimation → percentages, ratios — percentage complement

Q.54 [MCQ • Moderate]

Two successive discounts of 20% and 10% are equivalent to a single discount of

(A) 25% (B) 18% (C) 28% (D) 30%

Type / syllabus: Numerical computation & estimation → percentages, ratios — successive discounts; the 30% distractor is the intended trap

Q.55 [NAT • Moderate]

A town's population of 10,000 grows by 10% in one year and then falls by 10% in the next. The population at the end of two years is _____.

Type / syllabus: Numerical computation & estimation → percentages, ratios — successive percentage change on a base

Q.56 [MCQ • Hard]

An article sold at two-thirds of its usual selling price yields a loss of 10%. The profit percentage at the usual selling price is

(A) 15% (B) 30% (C) 50% (D) 35%

Type / syllabus: Numerical computation & estimation → percentages, ratios — two-step profit/loss; upper end of the GA 2-mark band

Q.57 [NAT • Moderate]

A sum of money is divided among A, B and C in the ratio $2 : 3 : 5$. If C receives Rs. 300 more than A, the total sum is _____.

Type / syllabus: Numerical computation & estimation → percentages, ratios — ratio with a difference constraint

Q.58 [MCQ • Hard]

In a test, 35% of the candidates failed in Mathematics, 45% failed in English and 20% failed in both. The percentage of candidates who passed in both subjects is

(A) 60% (B) 80% (C) 20% (D) 40%

Type / syllabus: Numerical computation & estimation → percentages, ratios — percentages with inclusion–exclusion

Q.59 [NAT • Hard]

Two varieties of rice costing Rs. 60 and Rs. 90 per kg are mixed in the ratio $2 : 1$ and the mixture is sold at Rs. 90 per kg. The profit percentage is _____ (rounded off to two decimal places).

Type / syllabus: Numerical computation & estimation → percentages, ratios — mixture combined with profit

Q.60 [MCQ • Hard]

If x is 25% more than y, then y is less than x by

(A) $33\frac{1}{3}\%$ (B) 15% (C) 25% (D) 20%

Type / syllabus: Numerical computation & estimation → percentages, ratios — percentage base reversal; the 25% distractor catches most candidates

---

Q.61 [MSQ • Moderate]

Which of the following statement(s) is/are correct?

(A) Two successive increases of 10% amount to a single increase of 21%

(B) A discount of 20% on a markup of 25% results in neither profit nor loss

(C) An increase of 50% followed by a decrease of 50% gives a net decrease of 20%

(D) 40% of 50 equals 50% of 40

Type / syllabus: Numerical computation & estimation → percentages, ratios — MSQ form; each option is a separate standard result

---

Q2. Averages, Mixtures & Alligation

10 questions • **+2 PYQ in Part B** • Official syllabus: Quantitative Aptitude → numerical computation and estimation; elementary statistics

Q.62

[NAT • Easy]

The average of 5, 10, 15, 20, 25 is _____.

Type / syllabus: Numerical computation & estimation → averages; elementary statistics — direct 1-mark computation

Q.63

[MCQ • Easy]

The average of the first ten natural numbers is

(A) 10 (B) 6 (C) 5.5 (D) 5

Type / syllabus: Numerical computation & estimation → averages; elementary statistics

Q.64

[NAT • Moderate]

The average of 20 numbers is 15. If the number 40 in the set is replaced by 20, the new average is _____.

Type / syllabus: Numerical computation & estimation → averages; elementary statistics — effect of replacing one observation

Q.65

[MCQ • Moderate]

The average age of 30 students is 14 years. When the teacher's age is included, the average becomes 15 years. The teacher's age, in years, is

(A) 15 (B) 40 (C) 45 (D) 44

Type / syllabus: Numerical computation & estimation → averages; elementary statistics — a GA staple; the 44 and 15 distractors are the two common slips

Q.66

[NAT • Moderate]

A vessel contains 40 litres of a milk–water mixture in the ratio 3 : 1. The quantity of water, in litres, to be added so that the ratio becomes 3 : 2 is _____.

Type / syllabus: Numerical computation & estimation → averages; elementary statistics — mixture with one component held fixed

Q.67

[MCQ • Moderate]

Two vessels each of 20 litres contain milk and water in the ratios 3 : 1 and 1 : 3 respectively. If both are emptied into a single vessel, the resulting ratio of milk to water is

(A) 1 : 1 (B) 3 : 1 (C) 1 : 3 (D) 2 : 3

Type / syllabus: Numerical computation & estimation → averages; elementary statistics — combining two mixtures

Q.68 [NAT • Moderate]

The average of six numbers is 30. The average of the first four is 25 and the average of the last three is 35. The fourth number is _____.

Type / syllabus: Numerical computation & estimation → averages; elementary statistics — overlapping groups; the overlap is the answer

---

Q.69 [MCQ • Hard]

Rice costing Rs. 30 per kg is mixed with rice costing Rs. 40 per kg to obtain a mixture worth Rs. 34 per kg. The two are mixed in the ratio

(A) 4 : 3 (B) 3 : 2 (C) 1 : 1 (D) 2 : 3

Type / syllabus: Numerical computation & estimation → averages; elementary statistics — rule of alligation

---

Q.70 [NAT • Hard]

The average weight of 10 persons increases by 1.5 kg when a person weighing 45 kg is replaced by a new person. The weight of the new person, in kg, is _____.

Type / syllabus: Numerical computation & estimation → averages; elementary statistics — replacement raises the total by $n\times$ the change in average

---

Q.71 [MSQ • Moderate]

A data set of five numbers has mean 20. Which of the following statement(s) is/are **necessarily** true?

(A) Adding 5 to every number makes the mean 25

(B) The median of the five numbers is 20

(C) The sum of the five numbers is 100

(D) Multiplying every number by 2 makes the mean 40

Type / syllabus: Numerical computation & estimation → averages; elementary statistics — properties of the mean under linear transformation

---

Q3. Time–Speed–Distance & Time–Work

14 questions • Official syllabus: Quantitative Aptitude → numerical estimation, ratios and rates

Q.72

[NAT • Easy]

A vehicle covers 120 km in 2 hours. Its average speed, in km/h, is _____.

Type / syllabus: Numerical estimation → rates, time–speed–distance, time–work

Q.73

[MCQ • Easy]

A speed of 72 km/h expressed in m/s is

(A) 18 (B) 36 (C) 25 (D) 20

Type / syllabus: Numerical estimation → rates, time–speed–distance, time–work — unit conversion; multiply by 5/18

Q.74

[NAT • Moderate]

A train 150 m long travelling at 54 km/h crosses a platform 250 m long in _____ seconds (rounded off to two decimal places).

Type / syllabus: Numerical estimation → rates, time–speed–distance, time–work — the train must cover its own length plus the platform

Q.75

[MCQ • Moderate]

A person travels from X to Y at 40 km/h and returns at 60 km/h. The average speed for the whole journey, in km/h, is

(A) 50 (B) 52 (C) 48 (D) 45

Type / syllabus: Numerical estimation → rates, time–speed–distance, time–work — harmonic mean; the 50 distractor is the arithmetic mean trap

Q.76

[NAT • Moderate]

A can complete a piece of work in 10 days and B in 15 days. Working together they complete it in _____ days.

Type / syllabus: Numerical estimation → rates, time–speed–distance, time–work — combined rates

Q.77

[MCQ • Moderate]

A and B together complete a work in 8 days, and A alone in 12 days. B alone would take

(A) 24 days (B) 4 days (C) 16 days (D) 20 days

Type / syllabus: Numerical estimation → rates, time–speed–distance, time–work

Q.78

[NAT • Moderate]

A boat travels at 10 km/h in still water and the stream flows at 2 km/h. The total time to travel 24 km downstream and return, in hours, is _____.

Type / syllabus: Numerical estimation → rates, time–speed–distance, time–work — boats and streams

Q.79 [MCQ • Moderate]

Two trains of lengths 100 m and 150 m travel towards each other at 40 km/h and 50 km/h. The time taken to cross each other completely is

(A) 9 s (B) 18 s (C) 25 s (D) 10 s

Type / syllabus: Numerical estimation → rates, time–speed–distance, time–work — relative speed in opposite directions

---

Q.80 [NAT • Hard]

A starts walking at 09:00 at 5 km/h. B starts along the same route at 10:00 at 7.5 km/h. Measured from B's start, B overtakes A after _____ hours.

Type / syllabus: Numerical estimation → rates, time–speed–distance, time–work — relative speed with a head start

---

Q.81 [NAT • Moderate]

Twelve workers build a wall in 10 days. Working at the same rate, 15 workers would build it in _____ days.

Type / syllabus: Numerical estimation → rates, time–speed–distance, time–work — inverse proportion

---

Q.82 [MCQ • Hard]

One tap fills a tank in 6 hours while another empties the full tank in 9 hours. With both open on an empty tank, the tank fills in

(A) 15 hours (B) 3.6 hours (C) it never fills (D) 18 hours

Type / syllabus: Numerical estimation → rates, time–speed–distance, time–work — pipes and cisterns with an outlet

---

Q.83 [NAT • Moderate]

A man covers 1 km in 12 minutes. His speed in km/h is _____.

Type / syllabus: Numerical estimation → rates, time–speed–distance, time–work

---

Q.84 [NAT • Hard]

A completes one-third of a job in 5 days and B completes two-fifths of the same job in 10 days. Working together they would complete the whole job in _____ days (rounded off to two decimal places).

Type / syllabus: Numerical estimation → rates, time–speed–distance, time–work — convert partial work into a rate first

---

Q.85 [MSQ • Moderate]

A train passes a stationary pole in 10 seconds and a 200 m platform in 20 seconds. Which of the following statement(s) is/are correct?

(A) The speed of the train is 72 km/h (B) The speed of the train is 20 m/s

(C) The length of the train is 200 m (D) The length of the train is 100 m

Type / syllabus: Numerical estimation → rates, time–speed–distance, time–work — two equations in two unknowns, posed as an MSQ

Q4. Number Systems, Powers, Logarithms & Series

12 questions • **+6 PYQ in Part B** • Official syllabus: Quantitative Aptitude → powers, exponents and logarithms, series

Q.86 [MCQ • Easy]

The value of $\log_{10} 1000$ is

(A) 3 (B) 1 (C) 10 (D) 100

Type / syllabus: Numerical computation → powers, exponents, logarithms, series

Q.87 [NAT • Easy]

The value of 2^{10} is _____.

Type / syllabus: Numerical computation → powers, exponents, logarithms, series

Q.88 [MCQ • Moderate]

If $\log_2 x = 5$, then x equals

(A) 32 (B) 10 (C) 25 (D) 64

Type / syllabus: Numerical computation → powers, exponents, logarithms, series

Q.89 [NAT • Moderate]

The sum of the first 20 natural numbers is _____.

Type / syllabus: Numerical computation → powers, exponents, logarithms, series — arithmetic series

Q.90 [MCQ • Moderate]

Find the next term: 1, 4, 9, 16, 25, _____

(A) 30 (B) 36 (C) 49 (D) 35

Type / syllabus: Numerical computation → powers, exponents, logarithms, series — series (perfect squares)

Q.91 [NAT • Moderate]

Find the next term: 2, 5, 11, 23, 47, _____

Type / syllabus: Numerical computation → powers, exponents, logarithms, series — series with the rule $a_{n+1} = 2a_n + 1$

Q.92 [MCQ • Moderate]

The highest common factor of 24 and 36 is

(A) 18 (B) 6 (C) 12 (D) 72

Type / syllabus: Numerical computation → powers, exponents, logarithms, series

Q.93 [NAT • Moderate]

The least common multiple of 12, 15 and 20 is _____.

Type / syllabus: Numerical computation → powers, exponents, logarithms, series

---

Q.94 [MCQ • Hard]

Given $\log_{10} 2 = 0.3010$, the number of digits in 2^{64} is

(A) 21 (B) 19 (C) 20 (D) 64

Type / syllabus: Numerical computation → powers, exponents, logarithms, series — logarithms used for digit counting; a recurring GA 2-mark idea

---

Q.95 [NAT • Hard]

The remainder when 7^{100} is divided by 5 is _____.

Type / syllabus: Numerical computation → powers, exponents, logarithms, series — modular arithmetic through cyclicity of the last digit

---

Q.96 [NAT • Moderate]

The sum of the infinite series $1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \cdots$ is _____.

Type / syllabus: Numerical computation → powers, exponents, logarithms, series — infinite geometric series

---

Q.97 [MSQ • Hard]

For $a, b > 0$ and a fixed base, which of the following identity/identities is/are correct?

(A) $\log_a a = 1$ (B) $\log(ab) = \log a + \log b$

(C) $\log(a + b) = \log a + \log b$ (D) $\log\left(\frac{a}{b}\right) = \log a - \log b$

Type / syllabus: Numerical computation → powers, exponents, logarithms, series — logarithm laws; option D is the standard misconception

---

Q5. Permutations, Combinations & Probability

11 questions • **+5 PYQ in Part B** • Official syllabus: Quantitative Aptitude → permutations and combinations, elementary probability

Q.98

[NAT • Easy]

The number of ways of arranging 4 distinct books on a shelf is _____.

Type / syllabus: Permutations and combinations; elementary probability

Q.99

[MCQ • Easy]

The value of $\binom{6}{2}$ is

(A) 20 (B) 30 (C) 15 (D) 12

Type / syllabus: Permutations and combinations; elementary probability

Q.100

[NAT • Moderate]

The number of distinct arrangements of the letters of the word **LEVEL** is _____.

Type / syllabus: Permutations and combinations; elementary probability — arrangements with repeated letters

Q.101

[MCQ • Moderate]

The number of ways of forming a committee of 3 from 7 people is

(A) 35 (B) 343 (C) 210 (D) 21

Type / syllabus: Permutations and combinations; elementary probability — the 210 distractor is the permutation count

Q.102

[NAT • Moderate]

A fair six-sided die is rolled once. The probability that the face shows a prime number is _____ (rounded off to two decimal places).

Type / syllabus: Permutations and combinations; elementary probability — elementary probability; $2, 3, 5$ are prime, 1 is not

Q.103

[MCQ • Moderate]

Two cards are drawn at random without replacement from a standard 52-card deck. The probability that both are kings is

(A) $\frac{4}{52}$ (B) $\frac{1}{169}$ (C) $\frac{1}{221}$ (D) $\frac{1}{13}$

Type / syllabus: Permutations and combinations; elementary probability

Q.104

[NAT • Moderate]

Using the digits $\{1, 2, 3, 4, 5\}$ without repetition, the number of three-digit numbers that can be formed is _____.

Type / syllabus: Permutations and combinations; elementary probability

Q.105 [MCQ • Hard]

The number of distinct ways in which 5 people can be seated around a circular table is

(A) 120 (B) 24 (C) 20 (D) 60

Type / syllabus: Permutations and combinations; elementary probability — circular permutations; 120 is the linear-arrangement trap

---

Q.106 [NAT • Hard]

A bag contains 4 red and 6 blue balls. Two balls are drawn at random without replacement. The probability that both are of the same colour is _____ (rounded off to three decimal places).

Type / syllabus: Permutations and combinations; elementary probability

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Q.107 [NAT • Moderate]

The number of diagonals in a regular decagon is _____.

Type / syllabus: Permutations and combinations; elementary probability — combinations applied to geometry

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Q.108 [MSQ • Hard]

Which of the following identity/identities is/are correct for $0 \le r \le n$?

(A) ${}^{n}P_{n} = \dfrac{n!}{2}$ (B) ${}^{n}P_{r} = \binom{n}{r} \cdot r!$

(C) $\binom{n}{0} = 1$ (D) $\binom{n}{r} = \binom{n}{n-r}$

Type / syllabus: Permutations and combinations; elementary probability — standard identities in MSQ form

---

Q6. Mensuration & Geometry

12 questions • **+3 PYQ in Part B** • Official syllabus: Quantitative Aptitude → mensuration and geometry

Q.109 [MCQ • Easy]

Taking $\pi = \frac{22}{7}$, the area of a circle of radius 7 units is

(A) 44 (B) 308 (C) 154 (D) 49

Type / syllabus: Mensuration and geometry — the 44 distractor is the circumference

Q.110 [NAT • Easy]

The perimeter of a square whose area is 144 square units is _____.

Type / syllabus: Mensuration and geometry

Q.111 [NAT • Moderate]

Taking $\pi = \frac{22}{7}$, the volume of a right circular cylinder of radius 7 and height 10 is _____.

Type / syllabus: Mensuration and geometry

Q.112 [MCQ • Moderate]

The total surface area of a cube of side 6 units is

(A) 144 (B) 252 (C) 216 (D) 36

Type / syllabus: Mensuration and geometry — 216 is also the volume, so read the question carefully

Q.113 [NAT • Moderate]

The angles of a triangle are in the ratio 2 : 3 : 4. The largest angle, in degrees, is _____.

Type / syllabus: Mensuration and geometry

Q.114 [MCQ • Moderate]

Each interior angle of a regular hexagon measures

(A) 108° (B) 120° (C) 135° (D) 60°

Type / syllabus: Mensuration and geometry

Q.115 [NAT • Moderate]

The area of an equilateral triangle of side 6 units is _____ (rounded off to two decimal places).

Type / syllabus: Mensuration and geometry

Q.116 [MCQ • Hard]

The volume of a sphere of radius 3 units is

(A) 9π (B) 12π (C) 27π (D) 36π

Type / syllabus: Mensuration and geometry

Q.117 [NAT • Hard]

The length of a rectangle is increased by 20% and its breadth is decreased by 20%. The percentage *decrease* in its area is _____.

Type / syllabus: Mensuration and geometry — mensuration combined with successive percentage change

Q.118 [NAT • Moderate]

A regular polygon has an exterior angle of 24°. Its number of sides is _____.

Type / syllabus: Mensuration and geometry — exterior angles of a polygon sum to 360°

Q.119 [MCQ • Hard]

A right circular cone and a right circular cylinder have the same radius and the same height. The ratio of the volume of the cone to that of the cylinder is

(A) 2 : 3 (B) 3 : 1 (C) 1 : 2 (D) 1 : 3

Type / syllabus: Mensuration and geometry

Q.120 [MSQ • Moderate]

For a right circular cylinder of radius 1 and height 2, which of the following statement(s) is/are correct?

(A) Its total surface area is 6π

(B) Its volume is 2π

(C) Its curved surface area is 4π

(D) Its volume is 4π

Type / syllabus: Mensuration and geometry — MSQ separating curved from total surface area

Section: Analytical Aptitude

A1. Logical Deduction & Syllogisms

12 questions • **+3 PYQ in Part B** • Official syllabus: Analytical Aptitude → logic: deduction and induction

Q.121 [MCQ • Easy]

All engineers are graduates. All graduates are literate. Which conclusion necessarily follows?

(A) Some graduates are not literate (B) No engineer is literate

(C) All engineers are literate (D) All literate people are engineers

Type / syllabus: Logic: deduction and induction — two-premise syllogism, GA 1-mark slot

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Q.122 [MCQ • Easy]

Some doctors are singers. All singers are artists. Which conclusion necessarily follows?

(A) No doctor is an artist (B) Some doctors are artists (C) All artists are doctors (D) All doctors are artists

Type / syllabus: Logic: deduction and induction — particular premise yields only a particular conclusion

---

Q.123 [MCQ • Moderate]

If it rains, the ground becomes wet. The ground is not wet. It follows that

(A) the ground is never wet (B) it did not rain

(C) it rained (D) it may or may not have rained

Type / syllabus: Logic: deduction and induction — modus tollens

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Q.124 [MCQ • Moderate]

Concluding *"P is true"* from the premises *"if P then Q"* and *"Q is true"* is

(A) the fallacy of affirming the consequent (B) a valid application of modus tollens

(C) the fallacy of denying the antecedent (D) a valid application of modus ponens

Type / syllabus: Logic: deduction and induction — named fallacies; increasingly used in GA critical-reasoning questions

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Q.125 [MSQ • Moderate]

From the premises *"All A are B"* and *"No B are C"*, which conclusion(s) necessarily follow(s)?

(A) Some C are A (B) No C are A (C) No A are C (D) All B are A

Type / syllabus: Logic: deduction and induction — MSQ syllogism, including the converse form

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Q.126 [MCQ • Moderate]

Only graduates are eligible to apply for the post. Ravi has applied and his application was accepted. It follows that

(A) Ravi is not a graduate (B) every graduate has applied

(C) Ravi is a graduate (D) nothing can be concluded about Ravi

Type / syllabus: Logic: deduction and induction — reading *only* as a necessary condition

Q.127 [MCQ • Moderate]

The statement *"All that glitters is not gold"* is logically equivalent to

(A) Only gold glitters (B) Everything that glitters is gold

(C) Nothing that glitters is gold (D) Some things that glitter are not gold

Type / syllabus: Logic: deduction and induction — scope of negation; a classic GA trap

Q.128 [MCQ • Hard]

No politician is honest. Some honest people are wealthy. Consider: **I.** Some wealthy people are not politicians. **II.** No wealthy person is a politician. Which follow(s)?

(A) Only II (B) Only I (C) Both I and II (D) Neither I nor II

Type / syllabus: Logic: deduction and induction — three-term syllogism with a particular premise

Q.129 [MCQ • Moderate]

All squares are rectangles. Which of the following must be true?

(A) No rectangle is a square (B) All rectangles are squares

(C) Some squares are not rectangles (D) Some rectangles are squares

Type / syllabus: Logic: deduction and induction — valid conversion of a universal affirmative

Q.130 [MSQ • Hard]

Which of the following argument form(s) is/are **valid**?

(A) From $P \to Q$ and P, conclude Q (B) From $P \to Q$ and $\neg P$, conclude $\neg Q$

(C) From $P \to Q$ and $Q \to R$, conclude $P \to R$ (D) From $P \to Q$ and $\neg Q$, conclude $\neg P$

Type / syllabus: Logic: deduction and induction — modus ponens, modus tollens, hypothetical syllogism vs. denying the antecedent

Q.131 [NAT • Moderate]

In a queue of 12 people, A is fifth from the front and B is sixth from the back. The number of people standing strictly between A and B is _____.

Type / syllabus: Logic: deduction and induction — positional reasoning

Q.132 [MCQ • Hard]

To determine X's present age, consider: **I.** X is five years older than Y. **II.** Y is 20 years old. Which is true?

(A) Statement II alone is sufficient

(B) Statement I alone is sufficient

(C) Even both statements together are not sufficient

(D) Both statements together are sufficient, but neither alone is sufficient

Type / syllabus: Logic: deduction and induction — data sufficiency format

A2. Blood Relations, Directions & Arrangements

12 questions • **+1 PYQ in Part B** • Official syllabus: Analytical Aptitude → numerical relations and reasoning

Q.133 [MCQ • Easy]

Pointing to a photograph, a man said, *"She is the daughter of my grandfather's only son."* The person in the photograph is the man's

(A) niece (B) aunt (C) sister (D) daughter

Type / syllabus: Numerical relations and reasoning → blood relations, directions, arrangements — blood relations; *grandfather's only son* resolves to the speaker's father

Q.134 [MCQ • Easy]

A person walks 5 km north, then 3 km east, then 5 km south. His displacement from the starting point is

(A) 13 km, towards the east (B) 3 km, towards the east

(C) zero (D) 5 km, towards the north

Type / syllabus: Numerical relations and reasoning → blood relations, directions, arrangements — direction sense

Q.135 [NAT • Moderate]

A person walks 4 km north and then 3 km east. His straight-line distance from the starting point, in km, is _____.

Type / syllabus: Numerical relations and reasoning → blood relations, directions, arrangements — direction sense with Pythagoras

Q.136 [MCQ • Moderate]

A is B's brother, C is A's mother and D is C's father. How is A related to D?

(A) Grandson (B) Father (C) Son (D) Nephew

Type / syllabus: Numerical relations and reasoning → blood relations, directions, arrangements — blood relations chain

Q.137 [MCQ • Moderate]

If $P \times Q$ means P *is the mother of* Q and $P + Q$ means P *is the father of* Q, then $A \times B + C$ means that A is C's

(A) mother (B) sister (C) grandmother (D) aunt

Type / syllabus: Numerical relations and reasoning → blood relations, directions, arrangements — symbolic relation coding, a standard GA 2-mark form

Q.138 [NAT • Moderate]

Seven people sit in a row. X is third from the left end and Y is third from the right end. The number of people sitting strictly between X and Y is _____.

Type / syllabus: Numerical relations and reasoning → blood relations, directions, arrangements — linear arrangement

Q.139 [MCQ • Moderate]

A man walks 10 m south, turns left and walks 5 m, then turns left again and walks 10 m. His distance from the starting point is

(A) zero (B) 15 m (C) 5 m (D) 25 m

Type / syllabus: Numerical relations and reasoning → blood relations, directions, arrangements — direction sense with turns

---

Q.140 [MCQ • Hard]

Five people P, Q, R, S, T sit in a row. Q sits immediately to the left of R; P sits between T and Q; S sits at the extreme left end. Who sits at the extreme right end?

(A) T (B) P (C) R (D) Q

Type / syllabus: Numerical relations and reasoning → blood relations, directions, arrangements — linear arrangement with adjacency and betweenness constraints

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Q.141 [NAT • Moderate]

In a class, Rahul is seventh from the top and eighteenth from the bottom in a merit list. The total number of students in the class is _____.

Type / syllabus: Numerical relations and reasoning → blood relations, directions, arrangements — ranking; the standard error is forgetting to subtract one

---

Q.142 [MCQ • Moderate]

Pointing to a man, a woman said, *"His mother is the only daughter of my mother."* How is the woman related to the man?

(A) Sister (B) Aunt (C) Mother (D) Grandmother

Type / syllabus: Numerical relations and reasoning → blood relations, directions, arrangements — blood relations

---

Q.143 [NAT • Hard]

A person walks 3 km east, then 4 km north, then 3 km west and finally 4 km north. His straight-line distance from the starting point, in km, is _____.

Type / syllabus: Numerical relations and reasoning → blood relations, directions, arrangements — direction sense; the east–west legs cancel

---

Q.144 [MSQ • Hard]

M is the son of N. N is the wife of O. O is the father of P. P is the sibling of Q, and M, P, Q are the only children of N and O. Which of the following statement(s) is/are correct?

(A) N is the mother of P (B) O is the husband of N

(C) M is the brother of P (D) Q is the daughter of O

Type / syllabus: Numerical relations and reasoning → blood relations, directions, arrangements — family tree in MSQ form; gender of Q is not given

---

A3. Coding–Decoding & Number/Letter Reasoning

10 questions • Official syllabus: Analytical Aptitude → analogy, numerical relations and reasoning

Q.145 [MCQ • Easy]

In a certain code **CAT** is written as **DBU**. In the same code **DOG** is written as

(A) FQI (B) EPH (C) CNF (D) EPG

Type / syllabus: Analogy; numerical relations and reasoning → coding–decoding, series — letter shift +1

Q.146 [NAT • Easy]

If $A = 1$, $B = 2$, …, $Z = 26$, then the sum of the letter values of the word **CAB** is _____.

Type / syllabus: Analogy; numerical relations and reasoning → coding–decoding, series — alphanumeric position coding

---

Q.147 [MCQ • Moderate]

If **PENCIL** is written as **QFODJM**, then **BOOK** is written as

(A) CQQM (B) CPPK (C) APNJ (D) CPPL

Type / syllabus: Analogy; numerical relations and reasoning → coding–decoding, series — letter shift +1

Q.148 [MCQ • Moderate]

Find the next term: A, C, F, J, O, _____

(A) T (B) S (C) V (D) U

Type / syllabus: Analogy; numerical relations and reasoning → coding–decoding, series — letter series with increasing gaps $2, 3, 4, 5, 6$

Q.149 [NAT • Moderate]

Find the next term: 3, 6, 12, 24, _____

Type / syllabus: Analogy; numerical relations and reasoning → coding–decoding, series — geometric series

---

Q.150 [NAT • Moderate]

If $2 * 3 = 13$ and $3 * 4 = 25$, then the value of $5 * 6$ is _____.

Type / syllabus: Analogy; numerical relations and reasoning → coding–decoding, series — inferring the operator rule $a * b = a^2 + b^2$

Q.151 [MCQ • Moderate]

Select the number that does **not** belong with the others.

(A) 27 (B) 8 (C) 100 (D) 125

Type / syllabus: Analogy; numerical relations and reasoning → coding–decoding, series — odd-one-out; the other three are perfect cubes

---

Q.152 [NAT • Hard]

In a certain code the value of a word is the sum of the positions of its letters in the alphabet, so that **MATH** has the value 42. The value of **LOGIC** is _____.

Type / syllabus: Analogy; numerical relations and reasoning → coding–decoding, series — alphanumeric coding, longer word

---

Q.153 [MCQ • Hard]

Find the next term: 5, 11, 23, 47, 95, _____

(A) 185 (B) 190 (C) 189 (D) 191

Type / syllabus: Analogy; numerical relations and reasoning → coding–decoding, series — series with the rule $a_{n+1} = 2a_n + 1$

Q.154 [MSQ • Hard]

Which of the following number(s) is/are divisible by 6?

(A) 411 (B) 726 (C) 234 (D) 132

Type / syllabus: Analogy; numerical relations and reasoning → coding–decoding, series — divisibility rules combined (2 and 3)

A4. Data Interpretation

14 questions • Official syllabus: Quantitative Aptitude → data interpretation: data graphs, bar/pie graphs, tables

Q.155 [NAT • Easy]

Refer to Table DI-1 below.

Product	2024	2025	2026	Total
P	20	25	30	75
Q	15	18	27	60
R	30	24	36	90
S	35	33	27	95
Total	100	100	120	320

Table DI-1: Units sold (in thousands) of four products over three years. The total number of units (in thousands) sold in 2026 is _____.

Type / syllabus: Data interpretation → tables, bar and pie charts — direct read-off

Q.156 [MCQ • Easy]

With reference to Table DI-1, the product with the highest sales in 2025 is

(A) Q (B) S (C) P (D) R

Type / syllabus: Data interpretation → tables, bar and pie charts — comparison within a column

Q.157 [NAT • Moderate]

With reference to Table DI-1, the percentage growth in the sales of product Q from 2024 to 2026 is _____.

Type / syllabus: Data interpretation → tables, bar and pie charts — percentage change across a row

Q.158 [MCQ • Moderate]

With reference to Table DI-1, the product whose sales declined in every successive year is

(A) S (B) Q (C) R (D) P

Type / syllabus: Data interpretation → tables, bar and pie charts — trend identification

Q.159 [NAT • Moderate]

With reference to Table DI-1, product R accounted for _____ % of the total units sold in 2026.

Type / syllabus: Data interpretation → tables, bar and pie charts — share of a column total

Q.160 [MCQ • Hard]

With reference to Table DI-1, the average annual sales of product P over the three years, in thousands of units, is

(A) 25 (B) 75 (C) 30 (D) 20

Type / syllabus: Data interpretation → tables, bar and pie charts — average across a row

Q.161 [MSQ • Hard]

With reference to Table DI-1, which of the following statement(s) is/are correct?

(A) In 2026, R sold more units than any other product

(B) The total sales in 2025 equal the total sales in 2024

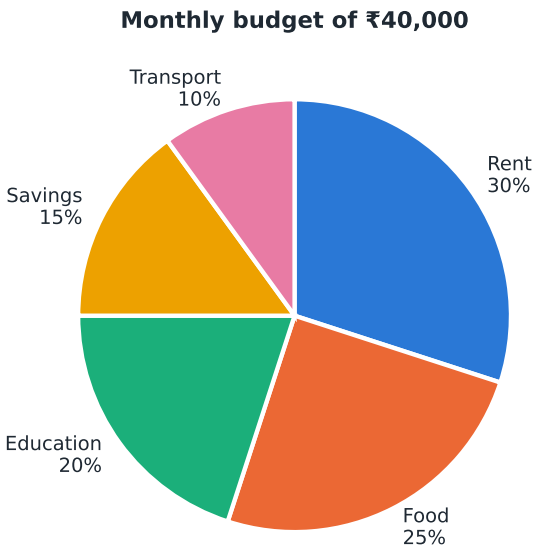
(C) S is the only product whose 2026 sales are below its 2024 sales

(D) The sales of Q more than doubled from 2024 to 2026

Type / syllabus: Data interpretation → tables, bar and pie charts — multi-cell reasoning in MSQ form

Q.162 [NAT • Easy]

A family's monthly budget of Rs. 40,000 is distributed as shown in the pie chart below.



The amount, in rupees, allocated to Rent is _____.

Type / syllabus: Data interpretation → tables, bar and pie charts — pie chart read-off

Q.163 [MCQ • Moderate]

For the budget pie chart above, the amount spent on Food exceeds that spent on Transport by

(A) Rs. 4,000 (B) Rs. 15,000 (C) Rs. 10,000 (D) Rs. 6,000

Type / syllabus: Data interpretation → tables, bar and pie charts — difference of two sectors

Q.164 [NAT • Moderate]

For the budget pie chart above, the central angle of the Education sector, in degrees, is _____.

Type / syllabus: Data interpretation → tables, bar and pie charts — converting a percentage to a sector angle

Q.165 [MCQ • Moderate]

If the monthly income rises to Rs. 50,000 and the same percentage split is retained, the amount saved would be

(A) Rs. 10,000 (B) Rs. 8,000 (C) Rs. 7,500 (D) Rs. 6,000

Type / syllabus: Data interpretation → tables, bar and pie charts — rescaling a distribution

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Q.166 [NAT • Hard]

For the budget pie chart above, Rent and Food together account for _____ % of the budget.

Type / syllabus: Data interpretation → tables, bar and pie charts — combining sectors

---

Q.167 [MCQ • Hard]

If the rent increases by 20% while every other amount stays unchanged, the new total monthly outgo, in rupees, is

(A) Rs. 44,000 (B) Rs. 42,400 (C) Rs. 48,000 (D) Rs. 42,000

Type / syllabus: Data interpretation → tables, bar and pie charts — percentage change applied to one component only

---

Q.168 [MSQ • Hard]

For the budget pie chart above, which of the following statement(s) is/are correct?

(A) The amount spent on Education is Rs. 8,000

(B) Transport spending is half the Education spending

(C) Rent is the single largest head of expenditure

(D) Savings exceed the Education allocation

Type / syllabus: Data interpretation → tables, bar and pie charts — multi-sector comparison in MSQ form

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Section: Spatial Aptitude

S1. Mirror Images, Water Images & Rotation

6 questions • **+1 PYQ in Part B** • Official syllabus: Spatial Aptitude → transformation of shapes: rotation, mirroring, translation

Q.169 [MCQ • Easy]

Which of the following words appears **unchanged** when held up to a vertical (left–right) mirror?

(A) PIP (B) BOB (C) DAD (D) MOM

Type / syllabus: Transformation of shapes → mirroring, rotation, translation — lateral inversion; every letter must have a vertical axis of symmetry *and* the word must be a palindrome

---

Q.170 [MCQ • Easy]

The word **AMBULANCE** is painted on the front of a vehicle so that a driver ahead reads it correctly in the rear-view mirror. As painted on the vehicle itself, the word appears

(A) with the letters in reverse order, each letter laterally inverted

(B) exactly as AMBULANCE

(C) with each letter turned upside down

(D) with the letters in reverse order but each letter upright

Type / syllabus: Transformation of shapes → mirroring, rotation, translation — the standard real-world lateral-inversion question

---

Q.171 [MSQ • Moderate]

Which of the following capital letters look(s) **unchanged** in a water image, i.e. a reflection in a horizontal line below the letter?

(A) A (B) C (C) D (D) B

Type / syllabus: Transformation of shapes → mirroring, rotation, translation — horizontal axis of symmetry, as distinct from the vertical axis

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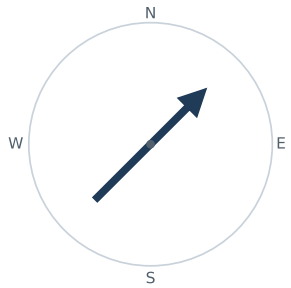
Q.172 [NAT • Moderate]

Starting from the 12 mark, the minute hand of a clock is rotated 210° clockwise. It now points at the _____ minute mark.

Type / syllabus: Transformation of shapes → mirroring, rotation, translation — rotation expressed in degrees; each minute mark is 6°

Q.173 [MCQ • Moderate]

The arrow shown below is rotated through 180° about the marked centre.



After the rotation the arrow points towards the

(A) south-west (B) north-east (C) north-west (D) south-east

Type / syllabus: Transformation of shapes → mirroring, rotation, translation — rotation of a directed figure about a point

---

Q.174 [NAT • Moderate]

On a standard die, opposite faces sum to 7. If the top face shows 2 and the front face shows 3, the face opposite the front shows _____.

Type / syllabus: Transformation of shapes → mirroring, rotation, translation — three-dimensional orientation of a cube

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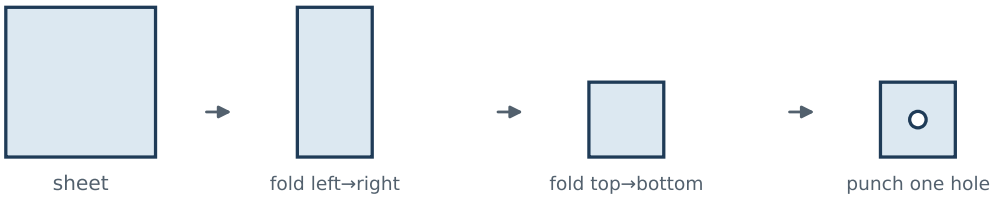
S2. Paper Folding, Cube Nets & Figure Counting

7 questions • **+1 PYQ in Part B** • Official syllabus: Spatial Aptitude → paper folding and cutting, patterns in 2 and 3 dimensions, assembling and grouping

Q.175

[NAT • Moderate]

A square sheet of paper is folded twice as shown and a single hole is then punched through all the layers.



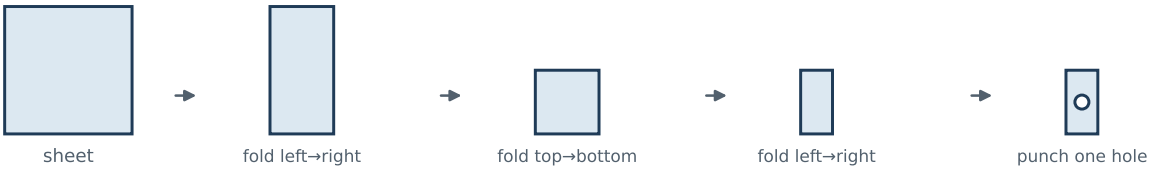
When the sheet is unfolded, the number of holes in it is _____.

Type / syllabus: Paper folding and cutting; patterns in 2 and 3 dimensions; assembling and grouping — paper folding and punching; each fold doubles the layer count

Q.176

[NAT • Hard]

A square sheet of paper is folded three times as shown and a single hole is then punched through all the layers.



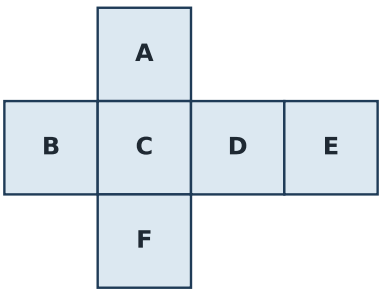
When the sheet is unfolded, the number of holes in it is _____.

Type / syllabus: Paper folding and cutting; patterns in 2 and 3 dimensions; assembling and grouping — three folds give 2^3 layers

Q.177

[MCQ • Moderate]

The figure below is the net of a cube.



When the net is folded into a cube, the face opposite to B is

(A) D (B) C (C) F (D) E

Type / syllabus: Paper folding and cutting; patterns in 2 and 3 dimensions; assembling and grouping — assembling a solid from its net; in a strip of four faces, opposite faces are two apart

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Q.178 [MCQ • Moderate]

For the same cube net, the face opposite to A is

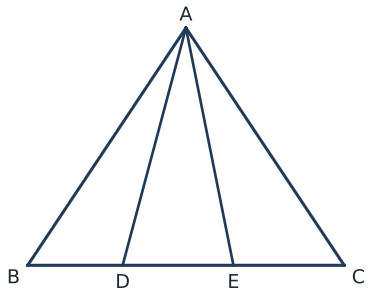
(A) C (B) E (C) B (D) F

Type / syllabus: Paper folding and cutting; patterns in 2 and 3 dimensions; assembling and grouping — the two faces attached above and below the same strip face are opposite

---

Q.179 [NAT • Moderate]

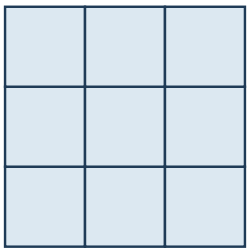
The number of triangles of any size in the figure below is _____.



Type / syllabus: Paper folding and cutting; patterns in 2 and 3 dimensions; assembling and grouping — figure counting; every pair of points on BC gives one triangle with apex A

Q.180 [NAT • Moderate]

The number of squares of any size in the figure below is _____.



Type / syllabus: Paper folding and cutting; patterns in 2 and 3 dimensions; assembling and grouping — figure counting; count 1×1, 2×2 and 3×3 squares separately

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Q.181 [NAT • Hard]

A solid cube of side 4 units is painted on all six faces and then cut into 64 unit cubes. The number of unit cubes having exactly two painted faces is _____.

Type / syllabus: Paper folding and cutting; patterns in 2 and 3 dimensions; assembling and grouping — patterns in three dimensions; the edge cubes excluding corners

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PART B — Previous Year Questions, topic-wise

Every question below is reproduced from an **official GATE question paper** published by the organising institute, and its answer is the **official answer key** for that paper. The source line under each question names the exact paper, question number and mark value. They are grouped under the same topic sets as Part A, so you can pair each practice set with the real thing.

Paper	PYQs used	Left out (figure-dependent)
GATE DA 2025	6	Q.3 image grid, Q.4 shapes, Q.8 dodecagon, Q.10 frequency chart
GATE DA 2024	6	Q.2 painting figure, Q.5 pie chart, Q.8 fund amounts, Q.9 dice net
GATE CS 2025 (Set 1)	7	Q.5 map, Q.7 diagram, Q.9 paper folding figure
GATE CS 2024 (Set 1)	8	Q.8 pie chart, Q.10 symmetry figure
GATE CS 2023	8	Q.2 marks-to-all, Q.10 transformation figure
GATE CS 2022	6	Q.4 statements, Q.5 plates, Q.9 triangle, Q.10 ropes figure
Total	**41**	

V1. Grammar, Tenses & Sentence Correction — previous year questions

3 previous-year questions • official papers, official answer keys • options are in the original paper order

Q.182 PYQ [MCQ • Easy]

Select the most appropriate option to complete the sentence.

We _____ tennis in the lawn when it suddenly started to rain.

(A) have been playing (B) had been playing (C) would have been playing (D) could be playing

Source: GATE DA 2025, General Aptitude Q.2 (1 mark) — official paper & official answer key

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Q.183 PYQ [MCQ • Easy]

Select the option with the correct sequence of articles to fill in the blanks.

Ravi had _____ younger brother who taught at _____ university. He was widely regarded as _____ honorable man.

(A) a; a; an (B) the; an; a (C) a; an; a (D) an; an; a

Source: GATE CS 2025 (Set 1), General Aptitude Q.1 (1 mark) — official paper & official answer key

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Q.184 PYQ [MCQ • Hard]

In the given text, the blanks are numbered (i)–(iv). Select the best match for all the blanks.

*Steve was advised to keep his head **(i)** before heading **(ii)** to bat; for, while he had a head **(iii)** batting, he could only do so with a cool head **(iv)** his shoulders.*

(A) (i) down (ii) down (iii) on (iv) for (B) (i) on (ii) down (iii) for (iv) on

(C) (i) down (ii) out (iii) for (iv) on (D) (i) on (ii) out (iii) on (iv) for

Source: GATE CS 2024 (Set 1), General Aptitude Q.6 (2 marks) — official paper & official answer key

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V2. Vocabulary, Idioms & Word Usage — previous year questions

6 previous-year questions • official papers, official answer keys • options are in the original paper order

Q.185 PYQ [MCQ • Easy]

*The CEO's decision to downsize the workforce was considered **myopic** because it sacrificed long-term stability to accommodate short-term gains.*

Select the most appropriate option that can replace the word **myopic** without changing the meaning of the sentence.

(A) visionary (B) shortsighted (C) progressive (D) innovative

Source: GATE CS 2025 (Set 1), General Aptitude Q.2 (1 mark) — official paper & official answer key

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Q.186 PYQ [MCQ • Easy]

Select the most appropriate pair of words.

The _____ is too high for it to be considered _____.

(A) fair / fare (B) faer / fair (C) fare / fare (D) fare / fair

Source: GATE CS 2022, General Aptitude Q.1 (1 mark) — official paper & official answer key

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Q.187 PYQ [MCQ • Easy]

If '→' denotes increasing order of intensity, then the meaning of the words [dry → arid → parched] is analogous to [diet → fast → _____]. Which one of the given options is appropriate to fill the blank?

(A) starve (B) reject (C) feast (D) deny

Source: GATE CS 2024 (Set 1), General Aptitude Q.1 (1 mark) — official paper & official answer key

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Q.188 PYQ [MCQ • Easy]

Select the most appropriate word.

We reached the station late, and _____ missed the train.

(A) near (B) nearly (C) utterly (D) mostly

Source: GATE CS 2023, General Aptitude Q.1 (1 mark) — official paper & official answer key

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Q.189 PYQ [MCQ • Moderate]

If '→' denotes increasing order of intensity, then the meaning of the words [sick → infirm → moribund] is analogous to [silly → _____ → daft]. Which option appropriately fills the blank?

(A) frown (B) fawn (C) vein (D) vain

Source: GATE DA 2024, General Aptitude Q.1 (1 mark) — official paper & official answer key

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Q.190 PYQ [MCQ • Moderate]

Column-I contains statements by Shanthala; Column-II contains responses by Kanishk. Identify the correct match.

Column-I	Column-II
P. This house is in a mess.	1. Alright, I won't bring it up during our conversations.
Q. I am not happy with the marks given to me.	2. Well, you can easily look it up.
R. Politics is a subject I avoid talking about.	3. No problem, let me clear it up for you.
S. I don't know what this word means.	4. Don't worry, I will take it up with your teacher.

(A) P–2; Q–3; R–1; S–4 (B) P–3; Q–4; R–1; S–2 (C) P–4; Q–1; R–2; S–3 (D) P–1; Q–2; R–4; S–3

Source: GATE DA 2025, General Aptitude Q.6 (2 marks) — official paper & official answer key

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V3. Reading Comprehension & Narrative Sequencing — previous year questions

7 previous-year questions • official papers, official answer keys • options are in the original paper order

Q.191 PYQ

[MCQ • Moderate]

"I put the brown paper in my pocket along with the chalks, and possibly other things. I suppose every one must have reflected how primeval and how poetical are the things that one carries in one's pocket: the pocket-knife, for instance, the type of all human tools, the infant of the sword. Once I planned to write a book of poems entirely about the things in my pocket. But I found it would be too long: and the age of the great epics is past."

(From G.K. Chesterton's *A Piece of Chalk*)

Based only on the information provided in the above passage, which one of the following statements is true?

(A) The author of the passage carries a mirror in his pocket to reflect upon things.

(B) The author of the passage had decided to write a poem on epics.

(C) The pocket-knife is described as the infant of the sword.

(D) Epics are described as too inconvenient to write.

Source: GATE CS 2025 (Set 1), General Aptitude Q.6 (2 marks) — official paper & official answer key

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Q.192 PYQ

[MCQ • Moderate]

Which one of the following sentence sequences creates a coherent narrative?

(i) Once on the terrace, on her way to her small room in the corner, she notices the man right away.
(ii) She begins to pant by the time she has climbed all the stairs.
(iii) Mina has bought vegetables and rice at the market, so her bags are heavy.
(iv) He was leaning against the parapet, watching the traffic below.

(A) (i), (ii), (iv), (iii) (B) (ii), (iii), (i), (iv) (C) (iv), (ii), (i), (iii) (D) (iii), (ii), (i), (iv)

Source: GATE CS 2023, General Aptitude Q.8 (2 marks) — official paper & official answer key

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Q.193 PYQ

[MCQ • Moderate]

The country of Zombieland is in distress since more than 75% of its working population is suffering from serious health issues. Studies conducted by competent health experts concluded that a complete lack of physical exercise among its working population was one of the leading causes of their health issues. As one of the measures to address the problem, the Government of Zombieland has decided to provide monetary incentives to those who ride bicycles to work.

Based only on the information provided above, which one of the following statements can be logically inferred with certainty?

(A) All the working population of Zombieland will henceforth ride bicycles to work.

(B) Riding bicycles will ensure that all of the working population of Zombieland is free of health issues.

(C) The health experts suggested to the Government of Zombieland to declare riding bicycles as mandatory.

(D) The Government of Zombieland believes that riding bicycles is a form of physical exercise.

Source: GATE CS 2023, General Aptitude Q.6 (2 marks) — official paper & official answer key

Q.194 PYQ [MCQ • Moderate]

Some people believe that "what gets measured, improves". Some others believe that "what gets measured, gets gamed". One possible reason for the difference in the beliefs is the work culture in organizations. In organizations with good work culture, metrics help improve outcomes. However, the same metrics are counterproductive in organizations with poor work culture.

Which one of the following is the CORRECT logical inference?

(A) Metrics are useful in organizations with poor work culture

(B) Metrics are useful in organizations with good work culture

(C) Metrics are always counterproductive in organizations with good work culture

(D) Metrics are never useful in organizations with good work culture

Source: GATE CS 2022, General Aptitude Q.6 (2 marks) — official paper & official answer key

Q.195 PYQ [MCQ • Moderate]

Thousands of years ago, some people began dairy farming. This coincided with a number of mutations in a particular gene that resulted in these people developing the ability to digest dairy milk.

Based on the given passage, which of the following can be inferred?

(A) All human beings can digest dairy milk.

(B) No human being can digest dairy milk.

(C) Digestion of dairy milk is essential for human beings.

(D) In human beings, the ability to digest dairy milk resulted from a mutated gene.

Source: GATE DA 2024, General Aptitude Q.6 (2 marks) — official paper & official answer key

Q.196 PYQ [MCQ • Moderate]

A survey for a certain year found that 90% of pregnant women received medical care at least once before giving birth. Of these women, 60% received medical care from doctors, while 40% received medical care from other healthcare providers.

Given this information, which one of the following statements can be inferred with certainty?

(A) More than half of the pregnant women received medical care at least once from a doctor.

(B) Less than half of the pregnant women received medical care at least once from a doctor.

(C) More than half of the pregnant women received medical care at most once from a doctor.

(D) Less than half of the pregnant women received medical care at most once from a doctor.

Source: GATE CS 2023, General Aptitude Q.4 (1 mark) — official paper & official answer key

Q.197 PYQ [MCQ • Hard]

In a recently conducted national entrance test, boys constituted 65% of those who appeared for the test. Girls constituted the remaining candidates and they accounted for 60% of the qualified candidates.

Which of the following statements is/are logically valid?

(A) Equal number of boys and girls qualified

(B) Equal number of boys and girls appeared for the test

(C) The number of boys who appeared was less than the number of girls who appeared

(D) The number of boys who qualified was less than the number of girls who qualified

Source: GATE CS 2022, General Aptitude Q.7 (2 marks) — official paper & official answer key

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V4. Verbal Analogies & Word Groups — previous year questions

1 previous-year question • official papers, official answer keys • options are in the original paper order

Q.198 PYQ

[MCQ • Easy]

Select the most appropriate option to complete the analogy.

Courage : Bravery :: Yearning : _____

(A) Longing (B) Yelling (C) Yawning (D) Glaring

Source: GATE DA 2025, General Aptitude Q.1 (1 mark) — official paper & official answer key

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Q1. Percentages, Ratio, Proportion, Profit & Loss — previous year questions

2 previous-year questions • official papers, official answer keys • options are in the original paper order

Q.199 PYQ

[MCQ • Moderate]

The number of coins of Rs. 1, Rs. 5 and Rs. 10 denominations that a person has are in the ratio $5 : 3 : 13$. Of the total amount, the percentage of money in Rs. 5 coins is

(A) 21% (B) $14\frac{2}{7}\%$ (C) 10% (D) 30%

Source: GATE CS 2024 (Set 1), General Aptitude Q.4 (1 mark) — official paper & official answer key

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Q.200 PYQ

[MCQ • Moderate]

A rectangle has length L and width W, where $L > W$. If the width, W, is increased by 10%, which one of the following statements is correct for all values of L and W?

(A) Perimeter increases by 10%

(B) Length of the diagonals increases by 10%

(C) Area increases by 10%

(D) The rectangle becomes a square

Source: GATE DA 2025, General Aptitude Q.5 (1 mark) — official paper & official answer key

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Q2. Averages, Mixtures & Alligation — previous year questions

2 previous-year questions • official papers, official answer keys • options are in the original paper order

Q.201 PYQ [MCQ • Moderate]

The average marks obtained by a class in an examination were calculated as 30.8. However, while checking the marks entered, the teacher found that the marks of one student were entered incorrectly as 24 instead of 42. After correcting the marks, the average becomes 31.4. How many students does the class have?

(A) 25 (B) 28 (C) 30 (D) 32

Source: GATE CS 2025 (Set 1), General Aptitude Q.3 (1 mark) — official paper & official answer key

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Q.202 PYQ [MCQ • Easy]

Consider the following sample of numbers:

9, 18, 11, 14, 15, 17, 10, 69, 11, 13

The median of the sample is

(A) 13.5 (B) 14 (C) 11 (D) 18.7

Source: GATE CS 2024 (Set 1), General Aptitude Q.3 (1 mark) — official paper & official answer key

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Q4. Number Systems, Powers, Logarithms & Series — previous year questions

6 previous-year questions • official papers, official answer keys • options are in the original paper order

Q.203 PYQ [MCQ • Moderate]

If $3^{x/2} = 27 \times 9^{x}$, then the value of $\dfrac{2^{x/2}}{(2^{x})^{2}}$ is

(A) 2^{-1} (B) 2^{0} (C) 2^{3} (D) 2^{15}

Source: GATE DA 2025, General Aptitude Q.9 (2 marks) — official paper & official answer key

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Q.204 PYQ [MCQ • Hard]

For positive non-zero real variables p and q, if

$$\log(p^2 + q^2) = \log p + \log q + 2\log 3,$$

then the value of $\dfrac{p^4 + q^4}{p^2 q^2}$ is

(A) 79 (B) 81 (C) 9 (D) 83

Source: GATE CS 2024 (Set 1), General Aptitude Q.5 (1 mark) — official paper & official answer key

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Q.205 PYQ [MCQ • Moderate]

A series of natural numbers $F_1, F_2, F_3, F_4, F_5, F_6, F_7, \ldots$ obeys $F_{n+1} = F_n + F_{n-1}$ for all integers $n \geq 2$. If $F_6 = 37$ and $F_7 = 60$, then what is F_1?

(A) 4 (B) 5 (C) 8 (D) 9

Source: GATE CS 2023, General Aptitude Q.3 (1 mark) — official paper & official answer key

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Q.206 PYQ [MCQ • Hard]

The sum of the following infinite series is

$$2+\frac{1}{2}+\frac{1}{3}+\frac{1}{4}+\frac{1}{8}+\frac{1}{9}+\frac{1}{16}+\frac{1}{27}+\cdots$$

(A) $\frac{11}{3}$ (B) $\frac{7}{2}$ (C) $\frac{13}{4}$ (D) $\frac{9}{2}$

Source: GATE DA 2024, General Aptitude Q.4 (1 mark) — official paper & official answer key

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Q.207 PYQ [MCQ • Hard]

Let r be a root of the equation $x^2+2x+6=0$. Then the value of the expression $(r+2)(r+3)(r+4)(r+5)$ is

(A) 51 (B) -51 (C) 126 (D) -126

Source: GATE CS 2022, General Aptitude Q.3 (1 mark) — official paper & official answer key

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Q.208 PYQ [MCQ • Moderate]

If two distinct non-zero real variables x and y are such that $(x+y)$ is proportional to $(x-y)$, then the value of $\frac{x}{y}$

(A) depends on xy (B) depends only on x and not on y

(C) depends only on y and not on x (D) is a constant

Source: GATE CS 2024 (Set 1), General Aptitude Q.2 (1 mark) — official paper & official answer key

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Q5. Permutations, Combinations & Probability — previous year questions

5 previous-year questions • official papers, official answer keys • options are in the original paper order

Q.209 PYQ [MCQ • Hard]

How many 4-digit positive integers divisible by 3 can be formed using only the digits $\{1,3,4,6,7\}$, such that no digit appears more than once in a number?

(A) 24 (B) 48 (C) 72 (D) 12

Source: GATE DA 2024, General Aptitude Q.3 (1 mark) — official paper & official answer key

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Q.210 PYQ [MCQ • Moderate]

The probability of a boy or a girl being born is $\frac{1}{2}$. For a family having three children, what is the probability of having two girls and one boy?

(A) $\frac{3}{8}$ (B) $\frac{1}{8}$ (C) $\frac{1}{4}$ (D) $\frac{1}{2}$

Source: GATE DA 2024, General Aptitude Q.7 (2 marks) — official paper & official answer key

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Q.211 PYQ [MCQ • Moderate]

A fair six-faced dice, with the faces labelled '1', '2', '3', '4', '5' and '6', is rolled thrice. What is the probability of rolling '6' exactly once?

(A) $\frac{75}{216}$ (B) $\frac{1}{6}$ (C) $\frac{1}{18}$ (D) $\frac{25}{216}$

Source: GATE CS 2025 (Set 1), General Aptitude Q.8 (2 marks) — official paper & official answer key

Q.212 PYQ [MCQ • Hard]

A shop has 4 distinct flavors of ice-cream. One can purchase any number of scoops of any flavor. The order in which the scoops are purchased is inconsequential. If one wants to purchase 3 scoops of ice-cream, in how many ways can one make that purchase?

(A) 4 (B) 20 (C) 24 (D) 48

Source: GATE CS 2025 (Set 1), General Aptitude Q.10 (2 marks) — official paper & official answer key

Q.213 PYQ [MCQ • Hard]

A box contains five balls of the same size and shape. Three of them are green coloured balls and two of them are orange coloured balls. Balls are drawn from the box one at a time. If a green ball is drawn, it is not replaced. If an orange ball is drawn, it is replaced with another orange ball. The first ball is drawn. What is the probability of getting an orange ball in the next draw?

(A) $\frac{1}{2}$ (B) $\frac{8}{25}$ (C) $\frac{19}{50}$ (D) $\frac{23}{50}$

Source: GATE CS 2022, General Aptitude Q.8 (2 marks) — official paper & official answer key

Q6. Mensuration & Geometry — previous year questions

3 previous-year questions • official papers, official answer keys • options are in the original paper order

Q.214 PYQ [MCQ • Hard]

A rectangular paper sheet of dimensions 54 cm × 4 cm is taken. The two longer edges of the sheet are joined together to create a cylindrical tube. A cube whose surface area is equal to the area of the sheet is also taken. Then, the ratio of the volume of the cylindrical tube to the volume of the cube is

(A) $\frac{1}{\pi}$ (B) $\frac{2}{\pi}$ (C) $\frac{3}{\pi}$ (D) $\frac{4}{\pi}$

Source: GATE CS 2024 (Set 1), General Aptitude Q.7 (2 marks) — official paper & official answer key

Q.215 PYQ [MCQ • Moderate]

Visualise two identical right circular cones such that one is inverted over the other and they share a common base. If a cutting plane passes through both the vertices of the resulting object, what shape does the outer boundary of the resulting cross-section make?

(A) A rhombus (B) A triangle (C) An ellipse (D) A hexagon

Source: GATE DA 2024, General Aptitude Q.10 (2 marks) — official paper & official answer key

Q.216 PYQ [MCQ • Moderate]

A function $y(x)$ is defined in the interval $[0, 1]$ as

$$y(x) = \begin{cases} 2, & 0 \le x < \frac{1}{3} \\ 3, & \frac{1}{3} \le x < \frac{3}{4} \\ 1, & \frac{3}{4} \le x \le 1 \end{cases}$$

Which one of the following is the area under the curve for the interval $[0, 1]$ on the x-axis?

(A) $\frac{5}{6}$ (B) $\frac{6}{5}$ (C) $\frac{13}{6}$ (D) $\frac{6}{13}$

Source: GATE CS 2022, General Aptitude Q.2 (1 mark) — official paper & official answer key

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A1. Logical Deduction & Syllogisms — previous year questions

3 previous-year questions • official papers, official answer keys • options are in the original paper order

Q.217 PYQ [MCQ • Moderate]

Consider two functions of time t: $f(t) = 0.01t^2$ and $g(t) = 4t$, where $0 < t < \infty$. Now consider the following two statements:

(i) For some $t > 0$, $g(t) > f(t)$.

(ii) There exists a T such that $f(t) > g(t)$ for all $t > T$.

Which one of the following options is TRUE?

(A) only (i) is correct (B) only (ii) is correct

(C) both (i) and (ii) are correct (D) neither (i) nor (ii) is correct

Source: GATE CS 2023, General Aptitude Q.7 (2 marks) — official paper & official answer key

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Q.218 PYQ [MCQ • Hard]

$f(x)$ and $g(y)$ are functions of x and y, respectively, and $f(x) = g(y)$ for all real values of x and y. Which one of the following options is necessarily TRUE for all x and y?

(A) $f(x) = 0$ and $g(y) = 0$

(B) $f(x) = g(y) =$ constant

(C) $f(x) \neq$ constant and $g(y) \neq$ constant

(D) $f(x) + g(y) - f(x) - g(y) = 0$

Source: GATE CS 2023, General Aptitude Q.9 (2 marks) — official paper & official answer key

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Q.219 PYQ [MCQ • Hard]

The weight of a person is 50 kg at birth and increases monotonically with age until the age of 50 years, after which it decreases monotonically. Two brothers, whose weights follow identical such functions of age, are of ages a_1 and a_2 with $a_1 < a_2$, and they have the same weight. Which one of the following is correct?

(A) $a_1 < a_2 < 50$ (B) $a_1 < 50 < a_2$

(C) $50 < a_1 < a_2$ (D) either $a_1 = 50$ or $a_2 = 50$

Source: GATE DA 2025, General Aptitude Q.7 (2 marks) — official paper & official answer key

A2. Blood Relations, Directions & Arrangements — previous year questions

1 previous-year question • official papers, official answer keys • options are in the original paper order

Q.220 PYQ

[MCQ • Moderate]

Consider the relationships among P, Q, R, S and T:

P is the brother of Q. S is the daughter of Q. T is the sister of S. R is the mother of Q.

The following statements are made based on the relationships given above:

(1) R is the grandmother of S. (2) P is the uncle of S and T.

(3) R has only one son. (4) Q has only one daughter.

Which one of the following options is correct?

(A) Both (1) and (2) are true.

(B) Both (1) and (3) are true.

(C) Only (3) is true.

(D) Only (4) is true.

Source: GATE CS 2025 (Set 1), General Aptitude Q.4 (1 mark) — official paper & official answer key

S1. Mirror Images, Water Images & Rotation — previous year questions

1 previous-year question • official papers, official answer keys • options are in the original paper order

Q.221 PYQ

[MCQ • Moderate]

Looking at the surface of a smooth three-dimensional object from the outside, which one of the following options is TRUE?

(A) The surface of the object must be concave everywhere.

(B) The surface of the object must be convex everywhere.

(C) The surface of the object may be concave in some places and convex in other places.

(D) The object can have edges, but no corners.

Source: GATE CS 2023, General Aptitude Q.5 (1 mark) — official paper & official answer key

S2. Paper Folding, Cube Nets & Figure Counting — previous year questions

1 previous-year question • official papers, official answer keys • options are in the original paper order

Q.222 PYQ

[MCQ • Hard]

A rectangular paper of 20 cm × 8 cm is folded 3 times. Each fold is made along the line of symmetry, which is perpendicular to its long edge. The perimeter of the final folded sheet (in cm) is

(A) 18 (B) 24 (C) 20 (D) 21

Source: GATE CS 2024 (Set 1), General Aptitude Q.9 (2 marks) — official paper & official answer key

Answer Key — Parts A and B

Answers only — no worked solutions, as requested. A key with several letters (e.g. **ABD**) is an MSQ, where every listed option must be selected and no other. For NAT questions any value within the usual GATE tolerance of the figure shown is accepted. The **Set** column is the topic set to revisit when you get one wrong; a • marks a previous-year question, whose answer is the official GATE key.

Q	Ans	Set	Q	Ans	Set	Q	Ans	Set	Q	Ans	Set
1	A	V1	**2**	B	V1	**3**	D	V1	**4**	C	V1
5	D	V1	**6**	B	V1	**7**	A	V1	**8**	D	V1
9	C	V1	**10**	B	V1	**11**	C	V1	**12**	ABC	V1
13	A	V1	**14**	C	V1	**15**	C	V2	**16**	B	V2
17	A	V2	**18**	A	V2	**19**	A	V2	**20**	C	V2
21	B	V2	**22**	A	V2	**23**	A	V2	**24**	ABD	V2
25	C	V2	**26**	D	V2	**27**	D	V2	**28**	D	V2
29	D	V3	**30**	C	V3	**31**	B	V3	**32**	C	V3
33	D	V3	**34**	A	V3	**35**	A	V3	**36**	A	V3
37	B	V3	**38**	C	V4	**39**	D	V4	**40**	A	V4
41	C	V4	**42**	C	V4	**43**	A	V4	**44**	ACD	V4
45	C	V4	**46**	A	Q1	**47**	200	Q1	**48**	C	Q1
49	20	Q1	**50**	B	Q1	**51**	5	Q1	**52**	C	Q1
53	300	Q1	**54**	C	Q1	**55**	9900	Q1	**56**	D	Q1
57	1000	Q1	**58**	D	Q1	**59**	28.57	Q1	**60**	D	Q1
61	ABD	Q1	**62**	15	Q2	**63**	C	Q2	**64**	14	Q2
65	C	Q2	**66**	10	Q2	**67**	A	Q2	**68**	25	Q2
69	B	Q2	**70**	60	Q2	**71**	ACD	Q2	**72**	60	Q3
73	D	Q3	**74**	26.67	Q3	**75**	C	Q3	**76**	6	Q3
77	A	Q3	**78**	5	Q3	**79**	D	Q3	**80**	2	Q3
81	8	Q3	**82**	D	Q3	**83**	5	Q3	**84**	9.38	Q3
85	ABC	Q3	**86**	A	Q4	**87**	1024	Q4	**88**	A	Q4
89	210	Q4	**90**	B	Q4	**91**	95	Q4	**92**	C	Q4
93	60	Q4	**94**	C	Q4	**95**	1	Q4	**96**	2	Q4
97	ABD	Q4	**98**	24	Q5	**99**	C	Q5	**100**	30	Q5
101	A	Q5	**102**	0.50	Q5	**103**	C	Q5	**104**	60	Q5
105	B	Q5	**106**	0.467	Q5	**107**	35	Q5	**108**	BCD	Q5
109	C	Q6	**110**	48	Q6	**111**	1540	Q6	**112**	C	Q6
113	80	Q6	**114**	B	Q6	**115**	15.59	Q6	**116**	D	Q6
117	4	Q6	**118**	15	Q6	**119**	D	Q6	**120**	ABC	Q6
121	C	A1	**122**	B	A1	**123**	B	A1	**124**	A	A1
125	BC	A1	**126**	C	A1	**127**	D	A1	**128**	B	A1
129	D	A1	**130**	ACD	A1	**131**	1	A1	**132**	D	A1
133	C	A2	**134**	B	A2	**135**	5	A2	**136**	A	A2
137	C	A2	**138**	1	A2	**139**	C	A2	**140**	C	A2
141	24	A2	**142**	C	A2	**143**	8	A2	**144**	ABC	A2
145	B	A3	**146**	6	A3	**147**	D	A3	**148**	D	A3
149	48	A3	**150**	61	A3	**151**	C	A3	**152**	46	A3
153	D	A3	**154**	BCD	A3	**155**	120	A4	**156**	B	A4
157	80	A4	**158**	A	A4	**159**	30	A4	**160**	A	A4
161	ABC	A4	**162**	12000	A4	**163**	D	A4	**164**	72	A4
165	C	A4	**166**	55	A4	**167**	B	A4	**168**	ABC	A4
169	D	S1	**170**	A	S1	**171**	BCD	S1	**172**	35	S1
173	A	S1	**174**	4	S1	**175**	4	S2	**176**	8	S2
177	A	S2	**178**	D	S2	**179**	6	S2	**180**	14	S2
181	24	S2	**182**	B	•V1	**183**	A	•V1	**184**	C	•V1
185	B	•V2	**186**	D	•V2	**187**	A	•V2	**188**	B	•V2
189	D	•V2	**190**	B	•V2	**191**	C	•V3	**192**	D	•V3
193	D	•V3	**194**	B	•V3	**195**	D	•V3	**196**	A	•V3
197	D	•V3	**198**	A	•V4	**199**	C	•Q1	**200**	C	•Q1
201	C	•Q2	**202**	A	•Q2	**203**	C	•Q4	**204**	A	•Q4
205	A	•Q4	**206**	B	•Q4	**207**	D	•Q4	**208**	D	•Q4
209	B	•Q5	**210**	A	•Q5	**211**	A	•Q5	**212**	B	•Q5
213	D	•Q5	**214**	A	•Q6	**215**	A	•Q6	**216**	C	•Q6
217	C	•A1	**218**	B	•A1	**219**	B	•A1	**220**	A	•A2
221	C	•S1	**222**	A	•S2						

Score sheet

Fill this in after checking the key. Score Part A and Part B separately — your Part B score is the one calibrated to the real exam.

Set	Topic	Part A	PYQ	A correct	PYQ correct
V1	Grammar, Tenses & Sentence Correction	14	3		
V2	Vocabulary, Idioms & Word Usage	14	6		
V3	Reading Comprehension & Narrative Sequencing	9	7		
V4	Verbal Analogies & Word Groups	8	1		
Q1	Percentages, Ratio, Proportion, Profit & Loss	16	2		
Q2	Averages, Mixtures & Alligation	10	2		
Q3	Time–Speed–Distance & Time–Work	14	–		
Q4	Number Systems, Powers, Logarithms & Series	12	6		
Q5	Permutations, Combinations & Probability	11	5		
Q6	Mensuration & Geometry	12	3		
A1	Logical Deduction & Syllogisms	12	3		
A2	Blood Relations, Directions & Arrangements	12	1		
A3	Coding–Decoding & Number/Letter Reasoning	10	–		
A4	Data Interpretation	14	–		
S1	Mirror Images, Water Images & Rotation	6	1		
S2	Paper Folding, Cube Nets & Figure Counting	7	1		
	Total	**181**	**41**		